A.J. INSTITUTE OF ENGINEERING & TECHNOLOGY, MANGALURU
Department of Master of Computer Applications (MCA) — Academic Year 2025–2026

SOFTWARE DESIGN & METHODOLOGY PRESENTATION

LegalAI: AI-Based Legal Assistant with Retrieval-Augmented Generation (RAG)

A Smart Legal Assistant to Answer Questions Accurately, Find Verified Lawyers, and Search Courts Across India

PRESENTER / STUDENT

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PROJECT GUIDE

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SOFTWARE DESIGN SPECIFICATION (SDS)

1. System Architecture: How the Parts Work Together

🏛️ The 4 Main Layers of the System

- **1. Web Front End (User Interface):** A clean, responsive website built with HTML, CSS, and JavaScript. Works smoothly on both mobile phones and laptops.
- **2. Backend Server (API Gateway):** Runs on Node.js/Flask. Manages logins, checks user permissions, and sends questions to the AI.
- **3. AI & Search Engine (RAG):** Finds exact law sections using FAISS vector search and generates plain-language answers using Google Gemini.
- **4. Database & Storage:** Safely stores user accounts, consultation bookings, and past chat records.

🧩 Why This Design is Simple & Flexible

- **Independent Parts (Modular):** You can upgrade the AI model or database anytime without changing the website code.
- **Fast and Responsive:** Only the lightweight web page runs in the user's browser, saving data and device battery.
- **Strict Security:** User passwords and chat records are kept private and separate from the public legal acts.
- **Never Shows a Blank Screen:** If the AI service is busy, built-in backup rules provide immediate answers so the system never fails.

Simple Flow: User asks a question on Web Page ⟶ Server checks security ⟶ FAISS finds matching law passages ⟶ Gemini writes an easy-to-read answer.

2. System Components & Database Design

⚙️ What Each Component Does

- **Controller:** Receives requests from the user, checks if they are logged in, and delivers the final answer.
- **RAG Service:** Takes the user's question, searches for relevant law paragraphs, and prepares context for the AI.
- **Retriever (FAISS):** A fast search tool that finds the 5 closest law sections matching the question in milliseconds.
- **Gemini Client:** Sends the law sections to Google Gemini to write a simple, easy-to-read explanation.
- **Database Manager:** Saves registered users, booked appointments, and past chat history.

💾 Database Tables Made Simple

- **Users Table:** Stores name, email, encrypted password, phone number, and account type (Client, Lawyer, Admin).
- **Documents & Chunks Table:** Stores official law names, section numbers, and small text pieces with duplicate prevention.
- **Chat Sessions Table:** Saves user questions and answers so clients can review past discussions anytime.
- **Consultations Table:** Stores booked lawyer appointments, chosen dates, time slots, and confirmation status.
- **Clean Storage:** Numbers and vectors stay in FAISS, while user records stay in the SQL database for maximum speed.

Zero Data Duplication: The database uses standard reference IDs (foreign keys), so if an advocate updates their contact number, it automatically updates across all appointments.

3. Step-by-Step Workflow: From Question to Answer

📚 Path A: Adding Legal Documents (Admin Workflow)

1. **Step 1: Document Upload —** Official Indian laws and court PDFs are loaded into the system by the admin.
2. **Step 2: Duplicate Check —** The system checks file fingerprints so the same document is never stored twice.
3. **Step 3: Clean Text —** Extra spaces, page numbers, and messy lines are removed automatically.
4. **Step 4: Break into Small Pieces (Chunks) —** Long law books are cut into small pieces (800 characters) for easy searching.
5. **Step 5: Save & Index —** Converted into smart numbers (embeddings) and saved in the FAISS vector index.

🔍 Path B: Asking a Legal Question (User Workflow)

1. **Step 1: User Types Question —** For example: "*What is the punishment for cheating?*"
2. **Step 2: Smart Search —** FAISS searches thousands of law sections and picks the top 5 most relevant pieces.
3. **Step 3: Quality Check (Relevance Gate) —** If no matching law is found, it honestly says "*No source found*" instead of guessing.
4. **Step 4: AI Writes Answer —** Google Gemini reads the found law pieces and explains them in simple English.
5. **Step 5: Display Sources & Disclaimer —** The screen shows the answer, the exact Act and Section, and an educational note.

Anti-Hallucination Guardrail: If someone asks an unrelated question like "*How to cook pasta?*", the system detects it is not law and politely refuses without wasting AI tokens.

EXPLANATION OF THE MODEL & METHODOLOGY

4. The AI Model: How It Understands Legal Questions

Text Embedder (MiniLM)

all-MiniLM-L6-v2

- Converts sentences into 384 smart numbers that capture true meaning.
- Understands synonyms: Knows that "*kicked out of house*" means "*eviction*".
- Small & lightweight: Runs in under 30 milliseconds on a normal laptop CPU.
- Fast and accurate without needing costly server hardware.

Fast Search Engine (FAISS)

FAISS IndexFlatIP

- Developed by Facebook AI to search millions of numbers in milliseconds.
- Uses exact cosine similarity to find closest matching law sections.
- Zero accuracy loss: Finds exact law sections reliably.
- Uses very little memory (only ~1.5 KB per law piece).

Answer Generator (Gemini)

Google Gemini 1.5

- Turns complex legal language into plain, easy-to-read sentences.
- Set to strict factual mode (Temperature 0.2) so it never makes up fake laws.
- Answers only from the provided law sections.
- Keeps explanations short, clear, and direct (capped at 600 tokens).

💡 Why We Picked This Combination

Older keyword search (like basic word matching) fails when users use everyday slang instead of formal legal Latin terms. Heavy AI models take too much computer memory and run slowly. Our combination gives **instant answers on a normal student laptop** while maintaining 100% legal accuracy.

EXPLANATION OF THE MODEL & METHODOLOGY

5. How Legal Text is Prepared & Guarded

🎀 Breaking Laws into Perfect Pieces (Chunking)

- **Why Not Search Whole Books?** Law books have hundreds of pages. Cutting them into smaller pieces lets the AI pinpoint exact sections.
- **Chunk Size (800 characters):** Just the right size to hold one complete legal sub-section and its explanation.
- **Overlap (100 characters):** Keeps a little bit of the previous piece so important sentences are not cut in half.
- **Keeps Important Words:** Words like "*must*", "*shall*", and "*not*" are never removed because they change whether an action is legal or illegal.

🛡️ Strict Safety Rules for the AI (Prompt Rules)

- **The Safety Guardrail:** If the search score is too low, the system stops and says "*No relevant law found*" to avoid fake answers.
- **Strict Instruction to Gemini:** "*Answer ONLY using the provided legal passages. If the passage does not have the answer, say you do not know.*"
- **Always Cite Sources:** The AI must always list the exact Act name, Section number, and year for every answer it gives.
- **Legal Disclaimer:** Every answer clearly reminds the user that this is for educational information, not professional legal counsel.

Fact-Checking Assurance: Because every answer is forced to display its source document, anyone can verify the law section directly in the official government gazette.

EXPLANATION OF THE MODEL & METHODOLOGY

6. Handling India's New Criminal Laws (BNS 2023)

⚖️ The Problem: India's Big Legal Overhaul in 2024

On July 1, 2024, India replaced its 160-year-old British-era laws (IPC, CrPC, Evidence Act) with new modern laws (BNS, BNSS, BSA). Common AI chatbots (like ChatGPT) still quote old, expired law sections (like IPC 302 or IPC 420). LegalAI includes a built-in **Law Transition Engine** that converts old sections to new active laws automatically!

Old Law (Pre-July 2024)

- **Murder / Killing:** Indian Penal Code (IPC) Section 302
- **Cheating & Fraud:** Indian Penal Code (IPC) Section 420
- **Anticipatory Bail:** Code of Criminal Procedure (CrPC) Section 438
- **High Court Inherent Powers:** CrPC Section 482
- **Mobile / Digital Proof:** Indian Evidence Act Section 65B

New Active Law in LegalAI (2024+)

- **Murder / Killing:** Bharatiya Nyaya Sanhita (BNS 2023) **Section 103**
- **Cheating & Fraud:** Bharatiya Nyaya Sanhita (BNS 2023) **Section 318(4)**
- **Anticipatory Bail:** BNSS 2023 **Section 482**
- **High Court Inherent Powers:** BNSS 2023 **Section 528**
- **Mobile / Digital Proof:** Bharatiya Sakshya Adhiniyam **Section 63**

Dual Benefit: If a citizen asks about "IPC 420", LegalAI cites **BNS Section 318(4)** as the active law, while still mentioning IPC 420 so old court precedents remain understandable.

7. The 6 Core Modules Built in the Project

1. User Login & Security

Fast user registration and login. Passwords encrypted with Bcrypt. Verifies 10-digit mobile numbers and manages user roles.

2. AI Legal Assistant

Understands user questions in plain words. Searches thousands of laws in milliseconds and writes simple, cited explanations.

3. Verified Lawyer Directory

Browse 50+ real Indian advocates. Includes State Bar Council license numbers, practice specialty, fees, and court experience.

4. Consultation Booking

Checks login before booking. Pick a date and time slot. Prevents double bookings and generates an instant unique Booking ID.

5. Courts Directory & Maps

Directory of 100+ Indian courts from Supreme Court to District Courts with live Google Maps directions and e-filing details.

6. Real-Time Data Sync

Automatic background sync. Any changes in the database immediately update spreadsheet files, and vice-versa.

System Status: All 6 core modules are fully built, tested, and working smoothly together in the project.

PERFORMANCE METRICS & EVALUATION

8. Test Results: Speed, Accuracy & Reliability

88.0%	0.82	94.2%	1.84s
RECALL SCORE	RANKING SCORE	ACCURACY SCORE	AVERAGE RESPONSE TIME
Target was ≥ 80% (Exceeded)	Correct law at Rank 1 or 2	Sentences backed by real law	Search takes 0.03s, AI writes in 1.8s

📊 What These Test Results Mean in Plain Words

- **Recall Score (88.0%):** In 88 out of 100 questions, the exact legal act and section were found in the top-5 search results. This easily beats our target of 80%.
- **Ranking Score (0.82):** When the search engine brings up law sections, the most important section appears at the very top (rank 1 or 2) almost every time.
- **Accuracy / Faithfulness (94.2%):** Tested against guide-reviewed questions, 94.2% of all statements produced were directly supported by real legal text with zero hallucinations.
- **Response Speed (1.84 seconds):** Searching thousands of laws takes only 32 milliseconds (0.03 seconds), and the full answer is generated in under 2 seconds, making it pleasant and responsive to use.

PERFORMANCE METRICS & EVALUATION

9. Safety Tests & Quality Checks

🧪 Tested with Real Expert Questions

- **Honest Testing:** Questions and answers were prepared directly from official Indian law books and verified by our project guide.
- **No Self-Testing Bias:** We did not let the AI test itself — answers were manually checked against real law texts.
- **Fair Data Split:** 40% of questions were used to tune settings, and 60% were kept hidden for final testing to make sure the AI truly understands.
- **Clean Database:** Duplicate checks removed repeated sentences so the search index stays clean and sharp.

🚫 Safety & Trick-Question Testing

- **Non-Legal Questions (e.g. 'How to bake a cake?'):** The search score was low, so the AI refused to answer and politely said: "*No relevant legal source found.*" (Zero tokens wasted).
- **Fake Laws (e.g. 'What is Section 999 IPC?'):** The system correctly recognized that Section 999 does not exist.
- **Hacking / Prompt Attacks:** Attempts to confuse the AI with trick instructions were safely blocked.
- **100% Core Test Pass:** User registration, lawyer search, appointment booking, and database sync all passed full testing.

Proven Stability: All 15 automated test cases passed without a single failure or server error.

PERFORMANCE METRICS & EVALUATION

10. Project Summary, Limitations & Future Plans

What We Achieved

- Built a complete legal helper that connects legal questions directly to real courts and verified lawyers.
- Stopped AI from making up fake laws using grounded search (RAG).
- Solved the transition from old IPC laws to new BNS 2023 laws.

Current Limitations

- Currently supports English only; Indian regional languages are not yet added.
- Old scanned paper judgments without searchable text need to be converted first.
- System is for educational help and does not replace hiring a human lawyer.

Future Plans

- **Voice Support:** Ask questions by voice in Kannada, Hindi, and Tamil.
- **Automatic Document Drafting:** Create rental agreements, RTI applications, and bail petitions automatically.
- **Live Court Case Tracking:** Connect directly to the Indian e-Courts portal to track live hearing dates.

⚖️ Mandatory Statutory & Ethical Disclaimer

LegalAI is an artificial intelligence legal research and educational assistant. It provides informational statutory analysis and does not constitute a formal advocate-client relationship or formal legal counsel under the Advocates Act, 1961.